IGARASHI Lab.
User Interface
Research Group

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Surfing the World

SmartLearn: Visual-Temporal Accessibility for Slide-based e-learning Videos

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Background

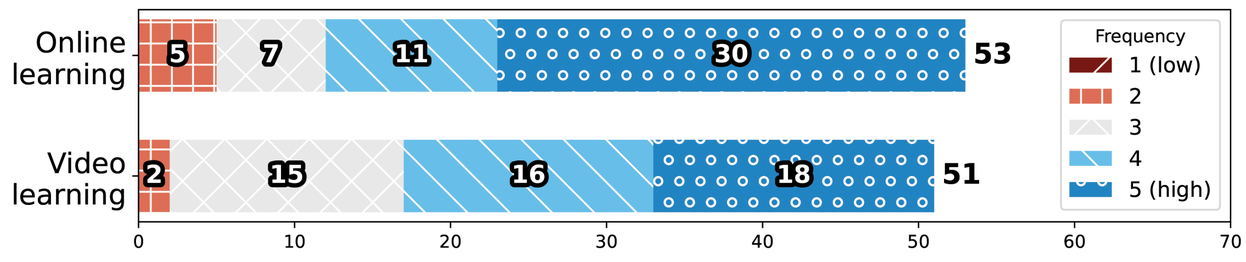
Videos are a very common e-learning content medium. But accessibility for videos is not straightforward. Videos present information temporally, vary in styles, and rely on both audio and visual understanding.

YouTube
coursera
udemy



Formative Study

Our formative study reveals many existing access barriers. We conducted a web-survey with 53 participants from various backgrounds (age, education, disability status, etc.). We see that even though video-based learning is widely used, many access barriers persist.



Key findings

Videos are still hard to...

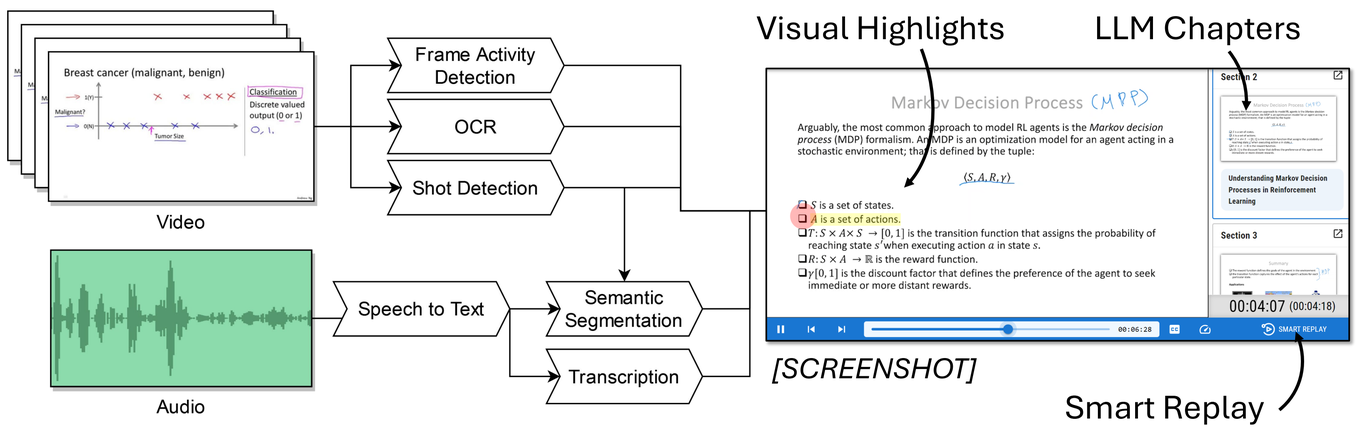
follow | see | navigate

What is SmartLearn?

An accessible video-based e-learning system utilizing multiple approaches to support learners with diverse abilities. Using temporal and visual accessibility solutions, such as:

Smart Replay	Visual Highlights	LLM Chapters
follow	see	navigate

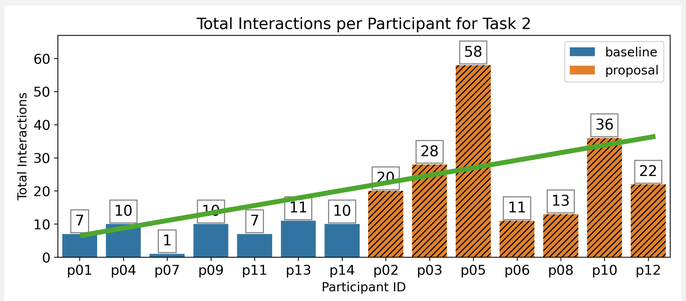
SmartLearn System



User Study

H1: SmartLearn is more **engaging**

More user interactions with the videos



H2: SmartLearn improves **learning**

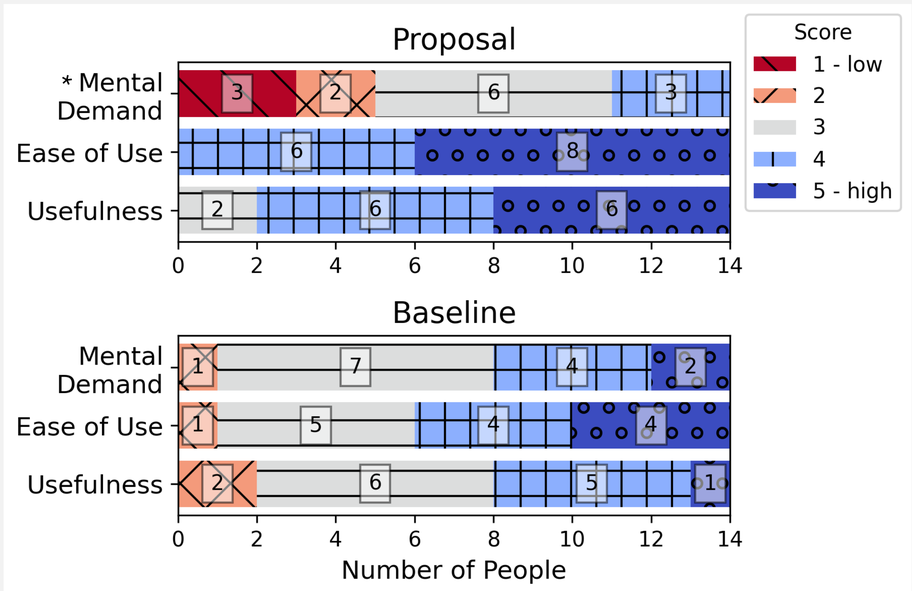
Visible increase in user quiz results

PID	Base	Proposal		PID	Base	Proposal	
P1	4	2		P8	4	3	
P2	4	3		P9	2	4	
P3	5	2		P10	3	5	
P4	2	2		P11	3	4	
P5	4	4		P12	4	2	
P6	2	3		P13	3	5	
P7	4	5		P14	4	4	

Comparing post-task quiz results:
- 6 improved score
- 3 maintained score
- 5 lowered score

H3: SmartLearn enhances **user experience**

Positive feedback in user evaluation



** Lower is better*

Lessons Learned

Chapter Generation Trade-off

More chapters give control, but less gives simpler experience. Leading to a required balance between **detail** and **simplicity**.

Replay Button Underutilization

Confusing distinction from chapters without unique enough function, and lack of transparency to the user on the outcome.

Speed Change Strategies

Words-per-second (WPS) lead to unnatural speech and confusing pace. But also, unintentionally contributed to increased attention.

Design Guidelines

- The design should be **responsive** to both user actions and content.
- The user experience should be **customizable** on all dimensions.
- The design should be **transparent** and communicates information about the system and content.

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